Tangent Line Approximation; Newton's Method

MATH 145 – Applied Calculus

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Local Linearization

Recall that:

Local Linearization

If $f(x)$ is differentiable at $x = a$, the **local linearization** (or *tangent line approximation*) to $f(x)$ near $x = a$ is the function

$$y = L(x) \approx f(a) + f'(a)(x - a)$$

If $f(x)$ is differentiable at $x = a$, you can approximate it by its tangent line.

Example: Linear Approximation

Example: Estimate $\sqrt{36.1}$ using linear approximation.

Example: Estimate $f(2.3)$ using $x = 2$ for $f(x) = x^3 e^{-x^2}$, if we know that $f'(x) = e^{-x^2}(3x^2 - 2x^4)$.

Example: Suppose that $g(x)$ is a function with $g(130) = 46$ and f'(130) $= 2$. Estimate $g(125.5)$.

Linear Approximation and Algebra

Question: Where does the linear approximation $l(x) = f(a) + f'(a)(x - a)$ intersect the x-axis?

Can we make this approximation better by moving the line? Can we *iterate* this?

This iterative algorithm is known as **Newton's Method**:

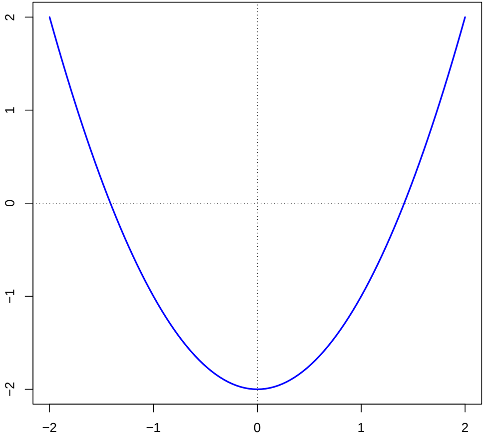
Newton's Method

Newton's Method for approximating solutions to $f(x) = 0$:

1. Use the graph to find a good starting point x_0 near to the solution x to $f(x) = 0$.
2. Find $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$.
3. Repeat until the difference $x_{n+1} - x_n$ is "small enough".

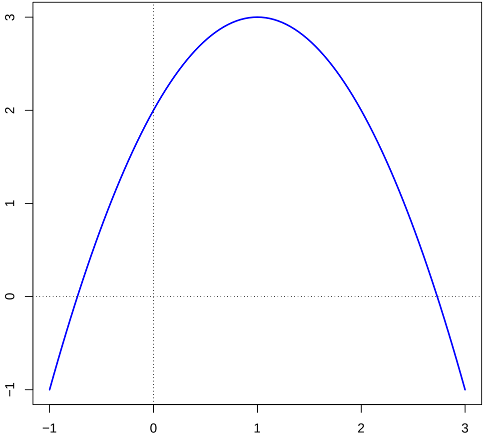
Linear Approximation and Algebra

Example: Estimate the roots of $f(x) = x^2 - 2$ using Newton's method.



Example: Newton's Method

Example: Estimate the roots of $f(x) = -x^2 + 2x + 2$ using Newton's method.



Example: Newton's Method

Example: Estimate the roots of $f(x) = x^3 - x^2 - 2x + 1$ using Newton's method.